

Karman Singh

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PROFESSIONAL SUMMARY

Software Engineer and IT undergraduate at NIT Jalandhar with hands-on experience in full-stack development (MERN), cloud infrastructure (AWS, Docker, Kubernetes), and machine learning. Proven ability to design and ship production-grade systems; seeking a software engineering internship to contribute to scalable, high-impact products.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, JavaScript, TypeScript, HTML, CSS, Java

Frontend Development: React.js, Responsive UI, REST API Integration, Component Design

Backend Development: Node.js, Express.js, RESTful APIs, JWT Authentication, AES-256 Encryption

Databases: MongoDB, SQL, Database Management Systems (DBMS)

Cloud & DevOps: AWS (ECS, EKS, CodePipeline, CodeBuild), Docker, Kubernetes, CI/CD Pipelines, GitHub Actions

Core CS Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Networks, OOPs, Software Design Patterns

Machine Learning : Machine Learning, Computer Vision, HuggingFace Transformers, Sentiment Analysis, Remote Sensing

Developer Tools: Git, GitHub, Linux, NLP, Agile, SDLC

EDUCATION

- **Dr. B.R. Ambedkar National Institute of Technology Jalandhar, Punjab**
Bachelor of Technology in Information Technology

Expected June 2027

WORK EXPERIENCE

- **Summer Software Intern - IIRS, ISRO** *June 2026 – July 2026*
 - Developing a Java 21 application to automate processing of Sentinel-2 Level-2A multispectral satellite imagery for large-scale vegetation analysis.
 - Implemented the MSAVI2 algorithm using Band 4 (Red) and Band 8 (NIR) and engineered a batch-processing pipeline to extract JPEG2000 spectral bands and export outputs in ENVI BSQ format.
 - Gained expertise in geospatial raster processing, scientific image analysis, and remote sensing data workflows.
- **Full Stack Developer - GDGC, NIT Jalandhar** *January 2025 - April 2026*
 - Developed and maintained full-stack features for HackMol, the official hackathon platform supporting event registrations, participant management, and live hackathon operations.
 - Built responsive React.js UI components and integrated RESTful APIs using Node.js and Express.js (MERN stack), improving platform user experience and reliability.
 - Performed debugging and performance optimization to ensure platform stability under high-traffic conditions during live events.

PROJECTS

- **MindMate - AI Mental Wellness Platform** *React, Node.js, MongoDB, AWS, Docker, HuggingFace* *June 2026* 🏆
 - Architected a full-stack mental wellness web app with AI-driven sentiment analysis and mood analytics using React, Node.js/Express, and MongoDB.
 - Designed an asynchronous, queue-based architecture to concurrently process journal submissions and HuggingFace Transformer inference, decoupling AI workloads from the main API.
 - Implemented JWT-based auth with refresh-token rotation and AES-256 field-level encryption for OWASP-compliant data privacy.
 - Containerized services with Docker, deployed to AWS via GitHub Actions CI/CD pipeline, and built a real-time content moderation pipeline for an anonymous community feature.

RELEVANT COURSEWORK

Data Structures and Algorithms — Operating Systems — Computer Networks — Database Management Systems — Design and Analysis of Algorithms — Object-Oriented Programming — Computer Organisation and Architecture — Machine Learning

EXTRACURRICULAR & LEADERSHIP

- **Core Team Member — GDG, TEDxNITJalandhar & Octaves**

NIT Jalandhar

- Contributed to organizing technical events, workshops, and cultural programs, demonstrating leadership, teamwork, and community engagement skills.

- **Handloom Hackathon 2025**

IIT Delhi

- One participant from the entire state of Punjab to clear the Selection Round.